



**CARINGBAH
HIGH
SCHOOL**

Name: _____

2018

**HSC
TRIAL**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided.
- In Questions 11–14, show relevant mathematical reasoning and/or calculations

Total marks: 70

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6–11)

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

Marker's Use Only

Question	1 – 10	11	12	13	14	Total	
Mark	/10	/15	/15	/15	/15	/70	%

This paper must not be removed from the examination room

Section I**10 marks****Attempt Questions 1–10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

1 Which expression is equal to $\sin x + \cos x$?

(A) $\sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$

(B) $\sqrt{2} \sin\left(x - \frac{\pi}{4}\right)$

(C) $2 \sin\left(x + \frac{\pi}{4}\right)$

(D) $2 \sin\left(x - \frac{\pi}{4}\right)$

2 What is the acute angle between the lines $3x + y = 4$ and $x - 2y = 5$?

(A) $\tan^{-1} 0.5$

(B) $\tan^{-1} 1$

(C) $\tan^{-1} 2$

(D) $\tan^{-1} 7$

3 Which of the polynomials are divisible by $x + 1$?

(I) $x^{2018} - 1$ (II) $x^{2017} - 1$ (III) $x^{2018} + 1$ (IV) $x^{2017} + 1$

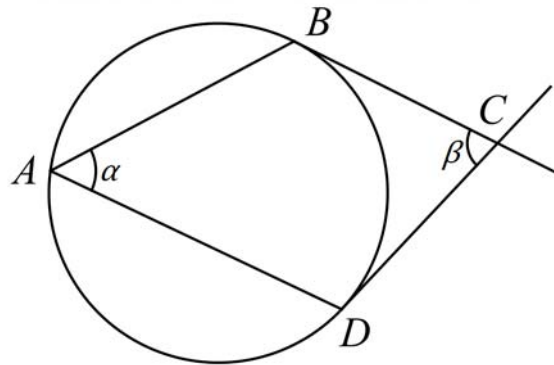
(A) (I) and (III) only

(B) (II) and (III) only

(C) (II) and (IV) only

(D) (I) and (IV) only

- 4 In the diagram below, if BC and DC are tangents then:



- (A) $\alpha + \beta = 180^\circ$
- (B) $2\alpha + \beta = 180^\circ$
- (C) $\alpha + 2\beta = 180^\circ$
- (D) $2\alpha + 2\beta = 180^\circ$
- 5 A particle is moving in simple harmonic motion which satisfies $\ddot{x} = -4x$.
- The particle starts from $x = 0$ with initial velocity 4 m/s.
- Which is a possible expression for the displacement, x , of the particle?
- (A) $x = 2 \sin 2t$
- (B) $x = 2 \sin 4t$
- (C) $x = 4 \sin 2t$
- (D) $x = 4 \sin 4t$

6 What is the derivative of $6\sin^{-1}\frac{x}{3}$?

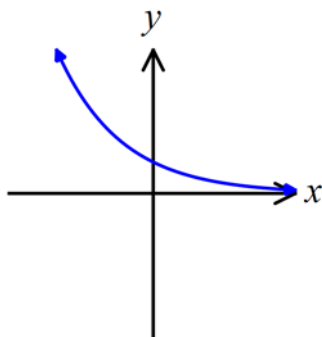
(A) $\frac{1}{\sqrt{9-x^2}}$

(B) $\frac{2}{\sqrt{9-x^2}}$

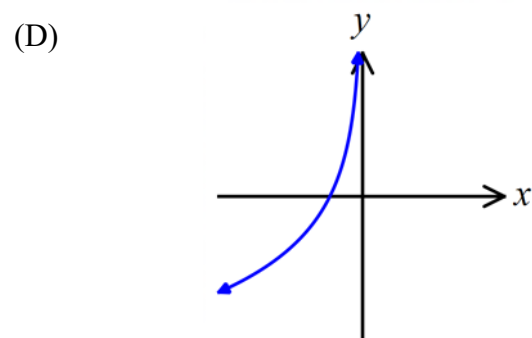
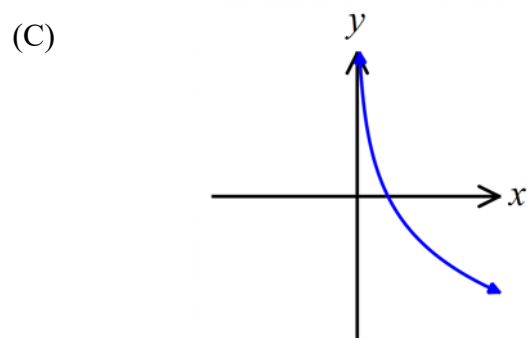
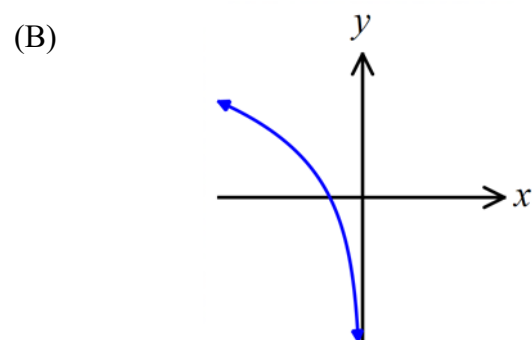
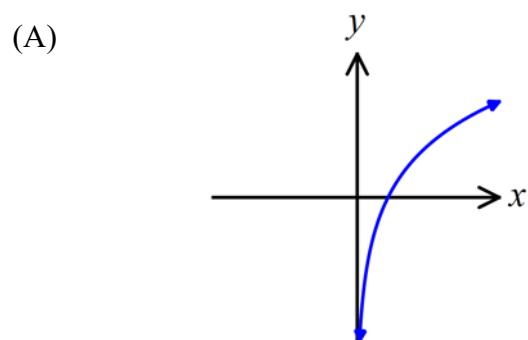
(C) $\frac{6}{\sqrt{9-x^2}}$

(D) $\frac{18}{\sqrt{9-x^2}}$

7 The diagram shows the graph of $y = f(x)$.



Which diagram shows the graph of $y = f^{-1}(x)$?



- 8 What are the asymptotes of $y = \frac{x+3}{(x-2)(x+1)}$?
- (A) $y = 0, x = 2, x = -1$
- (B) $y = 0, x = -2, x = 1$
- (C) $y = 3, x = 2, x = -1$
- (D) $y = 3, x = -2, x = 1$
- 9 Which of the following are the coordinates of the point of intersection of the normal $x + 2py = 8ap^3 + 4ap$ and the parabola $x^2 = 4ay$?
- (A) $(2ap, ap^2)$
- (B) $(4ap, 4ap^2)$
- (C) $(-2ap, ap^2)$
- (D) $(-4ap, 4ap^2)$
- 10 A particle moves in a straight line. Its position at any time t is given by

$$x = 3 \cos 2t + 4 \sin 2t.$$

The acceleration in terms of x is:

- (A) $\ddot{x} = -2x$
- (B) $\ddot{x} = -3x$
- (C) $\ddot{x} = -4x$
- (D) $\ddot{x} = -5x$

Section II**60 marks****Attempt Questions 11–14****Allow about 1 hour and 45 minutes for this section**

Answer each question on SEPARATE writing paper. Extra writing booklets are available.

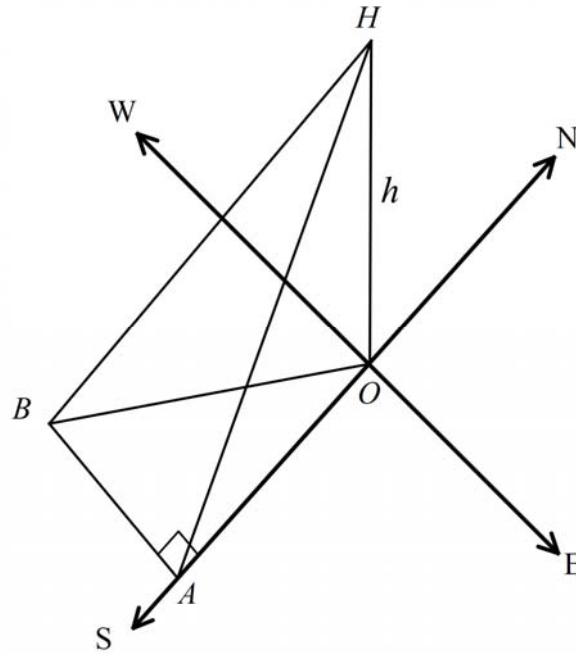
In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

	Marks
Question 11 (15 marks) Use SEPARATE writing paper.	
(a) A and B are the points $(1, 4)$ and $(5, 2)$ respectively. Find the coordinates of the point M which divides the interval AB externally in the ratio $2:3$.	2
(b) Solve $\frac{2}{x} \leq \frac{x}{2}$.	3
(c) Evaluate $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{4x}$.	2
(d) Water at a temperature of 24°C is placed in a freezer that maintains a constant temperature at -12°C . After time t minutes the rate of change of temperature T of the water is given by the formula: $\frac{dT}{dt} = -k(T + 12)$ where t is the time in minutes and k is a positive constant.	
(i) Show that $T = Ae^{-kt} - 12$ is a solution of this equation, where A is a constant.	1
(ii) Show that the value of A is 36.	1
(iii) After 15 minutes the temperature of the water falls to 9°C . Show that the value of k is 0.0359 correct to 3 significant figures.	1
(iv) Find to the nearest minute the time taken for the water to start freezing. (Freezing point of water is 0°C).	2
(e) (i) Factorise $e^{3x} + e^{3y}$.	1
(ii) If $e^x + e^y = 3$ and $e^{3x} + e^{3y} = 10$, find the exact value of $x + y$.	2

End of Question 11

Question 12 (15 marks) Use SEPARATE writing paper.

- (a) Point A is due south of the base of a hill, the angle of elevation from A to the top of the hill, H , is 46° . Another point B is due west of A and the angle of elevation from B to the top of the hill is 35° . The distance AB is 220 m.

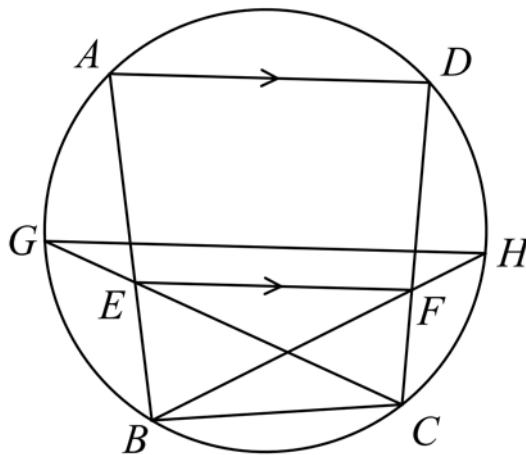


- (i) Show that $OA = h \tan 44^\circ$. 1
- (ii) Find the height of the hill, h correct to 1 decimal place. 2

Question 12 continues on page 8

Question 12 (continued)

- (b) In the diagram below $ABCD$ is a cyclic quadrilateral. E is a point on AB and F is a point on DC such that $EF \parallel AD$. BF produced and CE produced meet the circle through A, B, C, D at H and G respectively.



- (i) Show that $EBCF$ is a cyclic quadrilateral. 2
- (ii) Show that $GH \parallel EF$, giving reasons. 2
- (c) Let $f(x) = 2 \cos^{-1}\left(\frac{x}{3}\right)$.
- (i) State the domain and range of the function $f(x)$. 2
- (ii) Sketch the graph of $y = f(x)$ using $\frac{1}{3}$ of a page. 1
- (iii) Determine the inverse function $y = f^{-1}(x)$, and write down the domain of this inverse function. 2
- (iv) Hence or otherwise find the exact value of the volume of the solid of revolution formed when the region bounded by the curve $y = f(x)$, the x -axis and the y -axis is rotated about the y -axis. 3

End of Question 12

Marks

Question 13 (15 marks) Use SEPARATE writing paper.

- (a) A spherical metal ball is heated so that its diameter is increasing at a constant rate of 0.005 m/s. Given the surface area of the sphere is $S = \pi D^2$ where D is the diameter of the sphere, at what rate is the surface area of the metallic ball increasing when its diameter is 6 metres? Correct your answer to 2 decimal places. 2
- (b) The speed v (cm/s) of a particle moving in a straight line is given by $v^2 = 12 + 8x - 4x^2$, where the magnitude of its displacement from a fixed point O is x (cm).
- (i) Show that the motion is simple harmonic. 1
- (ii) Find the amplitude of the motion. 2
- (iii) Find the maximum speed of the particle and state where it occurs. 2
- (c) Let $f(x) = e^x + \ln x$.
- (i) Show that $f(x)$ is a monotonically increasing function for $x > 0$. Hence explain why $f(x)$ has an inverse. 2
- (ii) The graphs of $y = f(x)$ and $y = f^{-1}(x)$ meet at exactly one point P . Let α be the x -coordinate of P . Explain why α is a root of the equation 1
- $$e^x + \ln x - x = 0.$$
- (iii) Take 0.5 as a first approximation for α . Use one application of Newton's method to find a second approximation for α correct to 3 significant figures. 2
- (d) Prove by mathematical induction that, for $n \geq 1$, 3

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \frac{5}{3 \times 4 \times 2^3} + \cdots + \frac{n+2}{n(n+1)2^n} = 1 - \frac{1}{(n+1)2^n}$$

End of Question 13

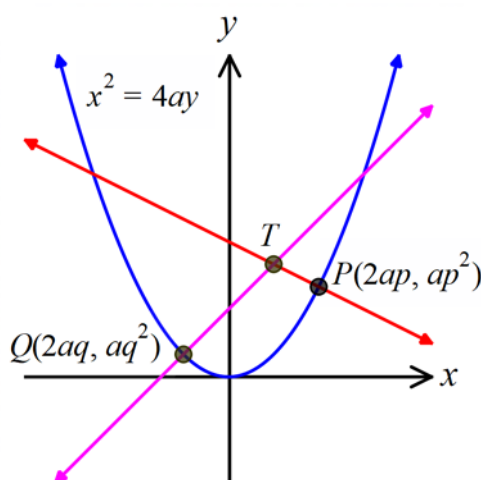
Question 14 (15 marks) Use SEPARATE writing paper.

- (a) Use the substitution $u = 4 - x^2$ to evaluate

3

$$\int_0^2 \frac{x}{\sqrt{4-x^2}} dx.$$

- (b) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are points on the parabola $x^2 = 4ay$. The normal to the parabola at P and Q intersect at the point $T(-apq(p+q), a(2+p^2+pq+q^2))$. (DO NOT prove this)



- (i) If $q = -3p$, show that the coordinates of T are $(-6ap^3, 2a + 7ap^2)$. 1
- (ii) There is a third normal to the parabola at the point $R(2ar, ar^2)$ that passes through the point T . Show that the parameter corresponding to the point R satisfies the equation 2

$$r^3 - 7p^2r + 6p^3 = 0.$$

- (iii) Hence find the coordinates of the point R in terms of p . 2

Question 14 continues on page 11

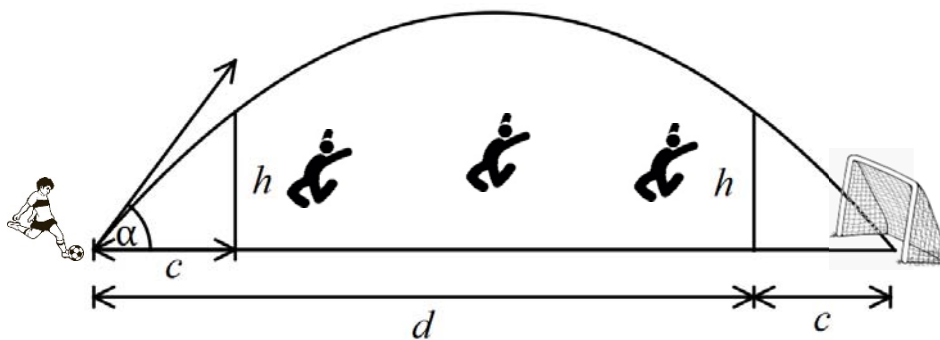
Question 14 (continued)

- (c) A soccer ball is at ground level on a horizontal plane. The ball was kicked at an angle α and with velocity v metres per second. You may assume that if the origin is taken to be the point of projection, the path of the ball is given by the equation

$$y = x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2v^2}$$

where $g \text{ ms}^{-2}$ is the acceleration due to gravity. (DO NOT prove this)

Steve Beckham wants to calculate the “perfect” angle to kick the ball to reach the goal. The defenders are standing between c metres and at most d metres from Steve. The defenders cannot jump higher than h metres. The trajectory of the ball is shown in the diagram below.



- (i) Explain why c and d are solutions to the equation

1

$$\frac{g \sec^2 \alpha}{2v^2} x^2 - x \tan \alpha + h = 0.$$

- (ii) Show that $c + d = \frac{2v^2 \tan \alpha}{g \sec^2 \alpha}$.

1

- (iii) Hence show that $\tan \alpha = \frac{h(c+d)}{cd}$.

2

- (iv) The horizontal range of the soccer ball is given by $R = \frac{v^2 \sin 2\alpha}{g}$. (DO NOT prove this)

3

Use the result in part (iii) to show $\frac{v^2}{g} = \frac{c^2 d^2 + h^2 R^2}{2hcd}$.

End of Exam

Section I

Q1	Q2	Q3	Q4	Q5
(A)	(D)	(D)	(B)	(A)
Q6	Q7	Q8	Q9	Q10
(C)	(C)	(A)	(B)	(C)

Question 1

$\sin x + \cos x = R \sin(x + \alpha)$ where

$$R = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \text{and} \quad \alpha = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4} \quad \therefore (A)$$

Question 2

$$3x + y = 4 \rightarrow m_1 = -3$$

$$x - 2y = 5 \rightarrow m_2 = \frac{1}{2}$$

$$\therefore \tan \theta = \left| \frac{-3 - \frac{1}{2}}{1 + (-3) \times \frac{1}{2}} \right| = 7$$

$$\theta = \tan^{-1}(7)$$

$\therefore (D)$

Question 3

factor $x + 1 \rightarrow x = -1$ is a root

$$(I) (-1)^{2018} - 1 = 0 \quad (IV) \quad (-1)^{2017} + 1 = 0$$

$\therefore (D)$

Question 4

Let O be the centre of circle

$\angle OBC = 90^\circ$ (radius $\perp$ tangent)

$\angle ODC = 90^\circ$ (radius $\perp$ tangent)

$\angle BOD = 2\alpha$

(angle at centre twice angle at circumference)

$$2\alpha + 90^\circ + 90^\circ + \beta = 360^\circ$$

(angle sum of quadrilateral $OBCD$)

$$\therefore 2\alpha + \beta = 180^\circ$$

$\therefore (B)$

Question 5

$$\ddot{x} = -4x \rightarrow n = 2.$$

$$x = a \sin 2t \rightarrow v = 2a \cos 2t,$$

$$t = 0, v = 4 \rightarrow 4 = 2a \cos 0 \rightarrow a = 2$$

$$\therefore x = 2 \sin 2t$$

$\therefore (A)$

Question 6

$$\frac{d}{dx} \left(6 \sin^{-1} \frac{x}{3} \right) = 6 \times \frac{1}{\sqrt{3^2 - x^2}} = \frac{6}{\sqrt{9 - x^2}} \quad \therefore (C)$$

Question 7

Reflect $y = f(x)$ about $y = x$.

$\therefore (C)$

Question 8

Vertical asymptote, denominator = 0

$$(x - 2)(x + 1) = 0 \rightarrow x = 2, x = -1$$

Horizontal asymptote,

$$y = \lim_{x \rightarrow \infty} \frac{x + 3}{(x - 2)(x + 1)} = \lim_{x \rightarrow \infty} \frac{x + 3}{x^2 - x - 2}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{x}{x^2} + \frac{3}{x^2}}{\frac{x^2}{x^2} - \frac{x}{x^2} - \frac{2}{x^2}} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x} + \frac{3}{x^2}}{1 - \frac{1}{x} - \frac{2}{x^2}}$$

$$= \frac{0 + 0}{1 - 0 - 0} = 0$$

$\therefore (A)$

Question 9

Point $(4ap, 4ap^2)$ lies on the equation of the normal by substitution into the equation of normal.

$$\text{LHS} = x + 2py$$

$$= 4ap + 2p \times 4ap^2$$

$$= 4ap + 8ap^3 = \text{RHS}$$

$\therefore (B)$

Question 10

$$x = 3 \cos 2t + 4 \sin 2t.$$

$$v = -6 \sin 2t + 8 \cos 2t$$

$$\ddot{x} = -12 \cos 2t - 16 \sin 2t$$

$$= -4(3 \cos 2t + 4 \sin 2t)$$

$$= -4x$$

$\therefore (C)$

Section II

Question 11

$$(a) \quad x = \frac{-3 \times 1 + 2 \times 5}{2 - 3} = -7, \quad y = \frac{-3 \times 4 + 2 \times 2}{2 - 3} = 8$$

$$\therefore M = (-7, 8)$$

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(b) $\frac{2}{x} \leq \frac{x}{2} \quad x \neq 0$

$$\frac{2}{x} \times x^2 \leq \frac{x}{2} \times x^2$$

$$2x \leq \frac{x^3}{2}$$

$$4x \leq x^3$$

$$0 \leq x^3 - 4x$$

$$0 \leq x(x-2)(x+2)$$

$$\therefore -2 \leq x < 0, x \geq 2$$

(c) $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{4x} = \frac{1}{4} \lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{x}$

$$= \frac{1}{4} \times \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{3}\right)}{\frac{x}{3}}$$

$$= \frac{1}{4} \times \frac{1}{3} \times 1 = \frac{1}{12}$$

(d) (i) $\frac{dT}{dt} = -kAe^{-kt}$

$$= -k(Ae^{-kt} - 12 + 12)$$

$$= -k(T + 12)$$

(ii) When $t = 0$, $T = 24^\circ$

$$24 = Ae^{-k \times 0} - 12$$

$$24 = A \times 1 - 12$$

$$A = 24 + 12 = 36$$

(iii) $9 = 36e^{-k \times 15} - 12$

$$21 = 36e^{-15k}$$

$$\frac{21}{36} = e^{-15k}$$

$$k = -\frac{1}{15} \ln\left(\frac{21}{36}\right) \approx 0.0359$$

(iv) $0 = 36e^{-kt} - 12$

$$12 = 36e^{-kt}$$

$$\frac{12}{36} = e^{-kt}$$

$$t = -\frac{1}{k} \ln\left(\frac{12}{36}\right)$$

$$\approx -\frac{1}{0.0359} \ln\left(\frac{12}{36}\right)$$

$$= 30.6... \approx 31 \text{ min}$$

(e) (i) $e^{3x} + e^{3y} = (e^x)^3 + (e^y)^3$

$$= (e^x + e^y)((e^x)^2 - e^x e^y + (e^y)^2)$$

(ii) $e^{3x} + e^{3y} = (e^x + e^y)((e^x)^2 - e^x e^y + (e^y)^2)$

$$10 = 3((e^x)^2 - e^x e^y + (e^y)^2)$$

$$10 = 3((e^x)^2 + 2e^x e^y + (e^y)^2 - 3e^x e^y)$$

$$10 = 3((e^x + e^y)^2 - 3e^x e^y)$$

$$10 = 3((3)^2 - 3e^{x+y})$$

$$e^{x+y} = \frac{17}{9}$$

$$x + y = \ln\left(\frac{17}{9}\right)$$

Question 12

(a) (i) $\frac{OA}{h} = \tan(90^\circ - 46^\circ)$

$$OA = h \tan 44^\circ$$

(ii) $(h \tan 55^\circ)^2 = 220^2 + (h \tan 44^\circ)^2$

$$h^2 (\tan^2 55^\circ - \tan^2 44^\circ) = 220^2$$

$$h = \frac{220}{\sqrt{\tan^2 55^\circ - \tan^2 44^\circ}}$$

$$= 209.09...$$

$$\approx 209.1$$

(b) (i) $\angle BEF = \angle EAD$ (corresponding angle on parallel lines, $AD \parallel EF$)

$$\angle EAD + \angle FCB = 180$$
 (Opposite angles of cyclic quad ABCD)
$$\therefore \angle BEF + \angle FCB = 180$$

$$\therefore EBCF \text{ is a cyclic quadrilateral (opposite angles are supplementary)}$$

(ii) $\angle GHF = \angle GCB$ (angles in the same segment, cyclic quad ABCD)

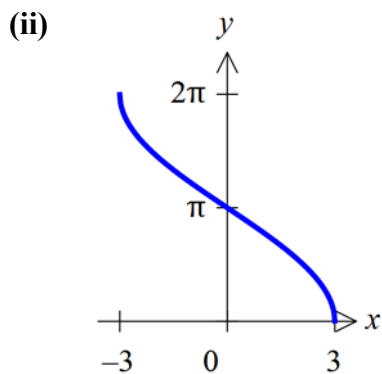
$$\angle EFB = \angle ECB$$
 (angles in the same segment, in cyclic quad EBCF)
$$\angle GHF = \angle EFB$$

$$GH \parallel EF$$

(corresponding angles are equal)

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(c) (i) $-3 \leq x \leq 3, 0 \leq y \leq 2\pi$



(iii) $x = 2 \cos^{-1}\left(\frac{y}{3}\right)$

$$\frac{x}{2} = \cos^{-1}\left(\frac{y}{3}\right)$$

$$\cos\left(\frac{x}{2}\right) = \frac{y}{3}$$

$$y = 3 \cos\left(\frac{x}{2}\right)$$

$$\therefore f^{-1}(x) = 3 \cos\left(\frac{x}{2}\right)$$

Domain: $0 \leq x \leq 2\pi$

(iv) $V = \pi \int_0^\pi x^2 dy$

$$= \pi \int_0^\pi \left(3 \cos\left(\frac{y}{2}\right)\right)^2 dy$$

$$= 9\pi \int_0^\pi \frac{1 + \cos y}{2} dy$$

$$= \frac{9\pi}{2} [y + \sin y]_0^\pi$$

$$= \frac{9\pi^2}{2}$$

Question 13

(a) $\frac{dS}{dt} = \frac{dS}{dD} \times \frac{dD}{dt}$

$$= 2\pi D \times \frac{dD}{dt}$$

$$= 2\pi \times 6 \times 0.005$$

$$\approx 0.19$$

(b) (i) $\frac{1}{2}v^2 = 6 + 4x - 2x^2$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 4 - 4x$$

$$\ddot{x} = -4(x - 1)$$

i.e. $\ddot{x}$ is proportional to x .

(ii) $v = 0 \rightarrow 12 + 8x - 4x^2 = 0$
 $-4(x^2 - 2x - 3) = 0$
 $(x - 3)(x + 1) = 0 \rightarrow x = 3, -1$
 Amplitude $= 3 - (-1) = 2$

(iii) Max speed occur at the centre of motion, at $x = 1$,
 $v^2 = 12 + 8(1) - 4(1)^2$
 $v^2 = 16 \rightarrow v_{\max} = 4 \text{ m/s}$

(c) (i) Since $e^x > 0$ and $\frac{1}{x}$ for $x > 0$,

$$\therefore f'(x) = e^x + \frac{1}{x} > 0 \text{ for } x > 0.$$

A monotonic increasing function implies 1 to 1 function as there are no turning points.

(ii) The two graphs intersect on the line $y = x$. So α will satisfy
 $f(x) = x \rightarrow e^x + \ln x = x$.
 Hence $e^x + \ln x - x = 0$

(iii) $x = 0.5 - \frac{e^{0.5} + \ln 0.5 - 0.5}{e^{0.5} + \frac{1}{0.5} - 1}$
 ≈ 0.328

(d) Prove $n = 1$ is true

$$LHS = \frac{3}{1 \times 2 \times 2} = \frac{3}{4} \quad RHS = 1 - \frac{1}{2 \times 2} = \frac{3}{4}$$

Therefore $n = 1$ is true.

Assume $n = k$ is true

$$\frac{3}{1 \times 2 \times 2} + \frac{4}{2 \times 3 \times 2^2} + \dots + \frac{k+2}{k(k+1)2^k} = 1 - \frac{1}{(k+1)2^k}$$

Prove $n = k + 1$ is true, i.e.

$$\frac{3}{1 \times 2 \times 2} + \dots + \frac{k+2}{k(k+1)2^k} + \frac{k+3}{(k+1)(k+2)2^{k+1}} = 1 - \frac{1}{(k+2)2^{k+1}}$$

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$$\begin{aligned}
 LHS &= \frac{3}{1 \times 2 \times 2} + \dots + \frac{k+2}{k(k+1)2^k} + \frac{k+3}{(k+1)(k+2)2^{k+1}} \\
 &= 1 - \frac{1}{(k+1)2^k} + \frac{k+3}{(k+1)(k+2)2^{k+1}} \\
 &= 1 - \left[\frac{2(k+2)}{(k+1)(k+2)2^{k+1}} - \frac{k+3}{(k+1)(k+2)2^{k+1}} \right] \\
 &= 1 - \frac{2(k+2) - k - 3}{(k+1)(k+2)2^{k+1}} \\
 &= 1 - \frac{k+1}{(k+1)(k+2)2^{k+1}} \\
 &= 1 - \frac{1}{(k+2)2^{k+1}} \\
 &= RHS
 \end{aligned}$$

$\therefore n = k + 1$ is true whenever $n = k$ is true.

$\therefore$ As it is true for $n = 1$, by induction, the result is true for $n \geq 1$.

Question 14

(a) $u = 4 - x^2 \rightarrow du = -2x dx$
 $x = 2 \rightarrow u = 0, \quad x = 0 \rightarrow u = 4$

$$\int_0^2 \frac{x}{\sqrt{4-x^2}} dx = -\frac{1}{2} \int_4^0 \frac{1}{\sqrt{u}} du$$

$$= \frac{1}{2} \int_0^4 u^{-\frac{1}{2}} du = \left[\sqrt{u} \right]_0^4 = 2 - 0 = 2$$

(b) (i) Substitute $q = -3p$
 $T(-ap(-3p)(p + -3p),$
 $\quad \quad \quad a(2 + p^2 + p(-3p) + (-3p)^2))$
 $= (3ap^2(-2p), a(2 + p^2 - 3p^2 + 9p^2))$
 $= (-6ap^3, 2a + 7ap^2)$

(ii) Equation of the normal at R is
 $x + ry = ar^3 + 2ar$.
 Since it passes through the point T
 The coordinates of T satisfies the above equations.
 $-6ap^3 + r(2a + 7ap^2) = ar^3 + 2ar$
 $-6ap^3 + 2ar + 7ap^2r = ar^3 + 2ar$
 $-6ap^3 + 7ap^2r = ar^3$
 $ar^3 - 7ap^2r + 6ap^3 = 0$
 $r^3 - 7p^2r + 6p^3 = 0$

(iii) The parameters of the three numbers satisfies the cubic equation in (ii).
 Sum of roots $= 0 \rightarrow p + -3p + r = 0$
 $-2p + r = 0 \rightarrow r = 2p$
 Coordinates of R is $(2a(2p), a(2p)^2)$
 Coordinates of R is $(4ap, 4ap^2)$

(c) (i) The projectile passes through the points (c, h) and (d, h)

$$h = x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2v^2}$$

$$\frac{gx^2 \sec^2 \alpha}{2v^2} - x \tan \alpha + h = 0$$

Therefore $x = c$ and $x = d$ are the roots of the quadratic equation.

(ii) Sum of the roots

$$\begin{aligned}
 c + d &= -(-\tan \alpha) \div \frac{g \sec^2 \alpha}{2v^2} \\
 &= \tan \alpha \times \frac{2v^2}{g \sec^2 \alpha} = \frac{2v^2 \tan \alpha}{g \sec^2 \alpha}
 \end{aligned}$$

(iii) Product of roots

$$cd = h \div \frac{g \sec^2 \alpha}{2v^2} = \frac{2v^2 h}{g \sec^2 \alpha}$$

Therefore using the result from (ii)

$$\begin{aligned}
 \frac{c+d}{cd} &= \frac{2v^2 \tan \alpha}{g \sec^2 \alpha} \div \frac{2v^2 h}{g \sec^2 \alpha} \\
 &= \frac{\cancel{2v^2} \tan \alpha}{g \sec^2 \alpha} \times \frac{g \cancel{\sec^2 \alpha}}{\cancel{2v^2} h} = \frac{\tan \alpha}{h}
 \end{aligned}$$

$$\therefore \tan \alpha = \frac{h(c+d)}{cd}$$

(iv) Rearranging $R = \frac{v^2 \sin 2\alpha}{g}$

$$\begin{aligned}
 v^2 &= \frac{Rg}{\sin 2\alpha} \\
 &= \frac{Rg}{2t} \quad (\text{where } t = \tan \alpha) \\
 &= \frac{Rg}{1+t^2} \\
 &= \frac{Rg}{2t} (1+t^2) \\
 &= \frac{Rg}{2} \left(\frac{1}{t} + t \right) \\
 &= \frac{Rg}{2} \left(\frac{cd}{h(c+d)} + \frac{h(c+d)}{cd} \right) \\
 &= \frac{Rg}{2} \left(\frac{cd}{hR} + \frac{hR}{cd} \right) \\
 &= \frac{\cancel{R}g}{2} \left(\frac{c^2 d^2 + h^2 R^2}{h \cancel{R} cd} \right) \\
 \frac{v^2}{g} &= \frac{c^2 d^2 + h^2 R^2}{2hcd}
 \end{aligned}$$